

GDGB: A Benchmark for Generative Dynamic Text-Attributed Graph Learning

GDGB (Peng et al., 2025) introduces a benchmark designed to evaluate **LLM-based generative models for dynamic text-attributed graphs (DyTAGs)**. It targets scenarios where nodes and edges evolve over time and are associated with rich textual information. The benchmark includes eight high-quality real-world DyTAG datasets and defines two novel generative tasks—**Transductive Dynamic Graph Generation (TDGG)** and **Inductive Dynamic Graph Generation (IDGG)**—for evaluating temporal, structural, and textual fidelity in graph generation.

LLM-Based DyTAG Generation Framework (With Examples)

GDGB proposes **GAG-General**, an LLM-based multi-agent framework for DyTAG generation. Each node is modeled as an agent, leveraging historical interaction memory and optionally a reflection mechanism to simulate evolution.

Example (Sephora Dataset):

A user node:

Name: Leonard
Skin Type: Dry
Skin Tone: Fair
Review Count: 5

A product node:

Name: Foundation
Brand: Lancôme
Loves: 5,314
Price: \$53

An edge (interaction):

Timestamp: 03.19.2023
Rating: 5
Review: "Nice color!"

GAG-General agents generate new interactions or even new nodes (IDGG). For example:

Generated User: Rebecca (Normal skin, Medium tan tone)
Generated Product: Lipstick (Brand: Sephora)
Generated Review at t' : Rating: 4, "Good price, but..."

TDGG vs. IDGG (With Examples)

TDGG: Generates new edges between existing nodes.

Example: From the known nodes (Leonard and Foundation), generate a new review edge at t' :

Review: "Bad experience!"

Rating: 1

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- **IDGG:** Generates both new nodes and edges.
Example: GAG-General introduces a new user and a new product, along with a review edge.

Multi-Faceted Evaluation Metrics

To evaluate generated DyTAGs, GDGB uses:

- **Graph Structural Metrics:** Degree and Spectral MMD, Power-law fit (e.g., $\alpha \in [2, 3]$)
- **Textual Quality Metrics:** Based on five LLM-evaluated criteria (e.g., Personality Depth, Content Richness)
- **Graph Embedding Metrics:** Joint evaluation of structure, time, and text using extended JL-Metric

Example (IMDB Dataset):

- **Structural Metric (IDGG):** Degree MMD = 0.145, $\alpha = 1.746$ (fails power-law validity)
- **Textual Quality (GPT + Memory + Reflection):** Score = 4.49
- **Embedding Score:** 0.563

Comparison with Other Models (With Examples)

Compared to baselines like VRDAG and DG-Gen, GAG-General:

- Generates more structurally faithful graphs (e.g., Degree MMD 0.073 on Cora vs. 0.877 for VRDAG)
- Generates semantically richer node/edge texts
- Retains graph-theoretic properties like hub node formation
Example: Generated hub product nodes include new items like "Vitamin E Moisturizer"

with realistic textual reviews

Why It Matters for the Survey

GDGB is the **first benchmark** to focus on **LLM-powered generation** of dynamic graphs with **rich textual attributes**. It provides datasets, tasks, metrics, and an LLM-based generation pipeline—all aligned with the core goals of dynamic/evolving graph learning. GAG-General also offers a modular, generalizable agent-based architecture, demonstrating how LLMs can simulate realistic dynamic graph growth across diverse domains. Thus, this work is essential for any survey on the intersection of LLMs and dynamic graph learning.